LAB ASSIGNMENT – 2
DATE – 19.01.2026

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ROLL NO. : UG/02/BTCSECSF/2023/003

SUBJECT : Unstructured Database Lab
SEC: E

Q1. Make a schema first and then insert 6 documents.

- Roll_no=[1,2,3,4,5,]
- Name=["Ram","Alex","John","Bob","Mukesh",”Danny”]
- Age=[20,19,40,55,30,28]
- Salary=[546.7,333.4,666.7,678.4,245.6,546.3]
- City=["A","B","C","D","E",”F”]
- Phone_no=[123,456,122,444,567,892]

```
collegeDB> db.employee.insertMany([
...   { Emp_id: 1, Name: "Ram", Age: 20, Salary: 546.7, City: "A", Phone_no: 123 },
...   { Emp_id: 2, Name: "Alex", Age: 19, Salary: 333.4, City: "B", Phone_no: 456 },
...   { Emp_id: 3, Name: "John", Age: 40, Salary: 666.7, City: "C", Phone_no: 122 },
...   { Emp_id: 4, Name: "Bob", Age: 55, Salary: 678.4, City: "D", Phone_no: 444 },
...   { Emp_id: 5, Name: "Mukesh", Age: 30, Salary: 245.6, City: "E", Phone_no: 567 },
...   { Emp_id: 6, Name: "Danny", Age: 28, Salary: 546.3, City: "F", Phone_no: 892 }
... ])
...
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('696dd7c4303ffd0e831e2621'),
    '1': ObjectId('696dd7c4303ffd0e831e2622'),
    '2': ObjectId('696dd7c4303ffd0e831e2623'),
    '3': ObjectId('696dd7c4303ffd0e831e2624'),
    '4': ObjectId('696dd7c4303ffd0e831e2625'),
    '5': ObjectId('696dd7c4303ffd0e831e2626')
  }
}
```

Q2. Write a query to update the name of RAM to SAM.

```
collegeDB> db.employee.updateOne({ Name: "Ram" }, { $set: {Name: "Sam"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q3. Write a query to display only the cities present in that collection.

```
collegeDB> db.employee.find({}, { _id: 0, City: 1 })
[
  { City: 'A' },
  { City: 'B' },
  { City: 'C' },
  { City: 'D' },
  { City: 'E' },
  { City: 'F' }
]
```

Q4. Write a query to update the salary by 10%.

```
collegeDB> db.employee.updateMany({}, { $mul: { Salary: 1.10 }})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 6,
  modifiedCount: 6,
  upsertedCount: 0
}
```

Q5. Write a query to display all the documents in ascending and descending order of age.

```
collegeDB> db.employee.find().sort({ Age: 1 })
[
  {
    _id: ObjectId('696dd7c4303ffd0e831e2622'),
    Emp_id: 2,
    Name: 'Alex',
    Age: 19,
    Salary: 366.74,
    City: 'B',
    Phone_no: 456
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2621'),
    Emp_id: 1,
    Name: 'Sam',
    Age: 20,
    Salary: 601.3700000000001,
    City: 'A',
    Phone_no: 123
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2626'),
    Emp_id: 6,
    Name: 'Danny',
    Age: 28,
    Salary: 600.93,
    City: 'F',
    Phone_no: 892
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2625'),
    Emp_id: 5,
    Name: 'Mukesh',
    Age: 30,
    Salary: 270.16,
    City: 'E',
    Phone_no: 567
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2623'),
    Emp_id: 3,
    Name: 'John',
    Age: 40,
    Salary: 733.3700000000001,
    City: 'C',
    Phone_no: 122
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2624'),
    Emp_id: 4,
    Name: 'Bob',
    Age: 55,
    Salary: 746.24,
    City: 'D',
    Phone_no: 444
  }
]
```

```
collegeDB> db.employee.find().sort({ Age: -1 })
[
  {
    _id: ObjectId('696dd7c4303ffd0e831e2624'),
    Emp_id: 4,
    Name: 'Bob',
    Age: 55,
    Salary: 746.24,
    City: 'D',
    Phone_no: 444
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2623'),
    Emp_id: 3,
    Name: 'John',
    Age: 40,
    Salary: 733.3700000000001,
    City: 'C',
    Phone_no: 122
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2625'),
    Emp_id: 5,
    Name: 'Mukesh',
    Age: 30,
    Salary: 270.16,
    City: 'E',
    Phone_no: 567
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2626'),
    Emp_id: 6,
    Name: 'Danny',
    Age: 28,
    Salary: 600.93,
    City: 'F',
    Phone_no: 892
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2621'),
    Emp_id: 1,
    Name: 'Sam',
    Age: 20,
    Salary: 601.3700000000001,
    City: 'A',
    Phone_no: 123
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2622'),
    Emp_id: 2,
    Name: 'Alex',
    Age: 19,
    Salary: 366.74,
    City: 'B',
    Phone_no: 456
  }
]
```

Q6. Write a query to display all the documents with City → A,B,C.

```
collegeDB> db.employee.find({ City: { $in: ["A","B","C"]}})
[
  {
    _id: ObjectId('696dd7c4303ffd0e831e2621'),
    Emp_id: 1,
    Name: 'Sam',
    Age: 20,
    Salary: 601.3700000000001,
    City: 'A',
    Phone_no: 123
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2622'),
    Emp_id: 2,
    Name: 'Alex',
    Age: 19,
    Salary: 366.74,
    City: 'B',
    Phone_no: 456
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2623'),
    Emp_id: 3,
    Name: 'John',
    Age: 40,
    Salary: 733.3700000000001,
    City: 'C',
    Phone_no: 122
  }
]
```

Q7. Write a query to display only two documents from the entire collection.

```
collegeDB> db.employee.find().limit(2)
[
  {
    _id: ObjectId('696dd7c4303ffd0e831e2621'),
    Emp_id: 1,
    Name: 'Sam',
    Age: 20,
    Salary: 601.3700000000001,
    City: 'A',
    Phone_no: 123
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2622'),
    Emp_id: 2,
    Name: 'Alex',
    Age: 19,
    Salary: 366.74,
    City: 'B',
    Phone_no: 456
  }
]
```

Q8. Write a query to delete a document with EMP_ID:5.

```
collegeDB> db.employees.deleteOne({ employee_id: 5 })
{ acknowledged: true, deletedCount: 1 }
```

Q9. Write a query to display all the documents with AGE greater than 20.

```
collegeDB> db.employee.find({ Age: { $gt: 20 } })
[
  {
    _id: ObjectId('696dd7c4303ffd0e831e2623'),
    Emp_id: 3,
    Name: 'John',
    Age: 40,
    Salary: 733.3700000000001,
    City: 'C',
    Phone_no: 122
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2624'),
    Emp_id: 4,
    Name: 'Bob',
    Age: 55,
    Salary: 746.24,
    City: 'D',
    Phone_no: 444
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2626'),
    Emp_id: 6,
    Name: 'Danny',
    Age: 28,
    Salary: 600.93,
    City: 'F',
    Phone_no: 892
  }
]
```

Q10. Write a query to display all the documents with AGE less than 20.

```
collegeDB> db.employee.find({ Age: { $lt: 20 } })
[
  {
    _id: ObjectId('696dd7c4303ffd0e831e2622'),
    Emp_id: 2,
    Name: 'Alex',
    Age: 19,
    Salary: 366.74,
    City: 'B',
    Phone_no: 456
  }
]
```

Q11. Write a query to display all the documents with AGE equals to 20.

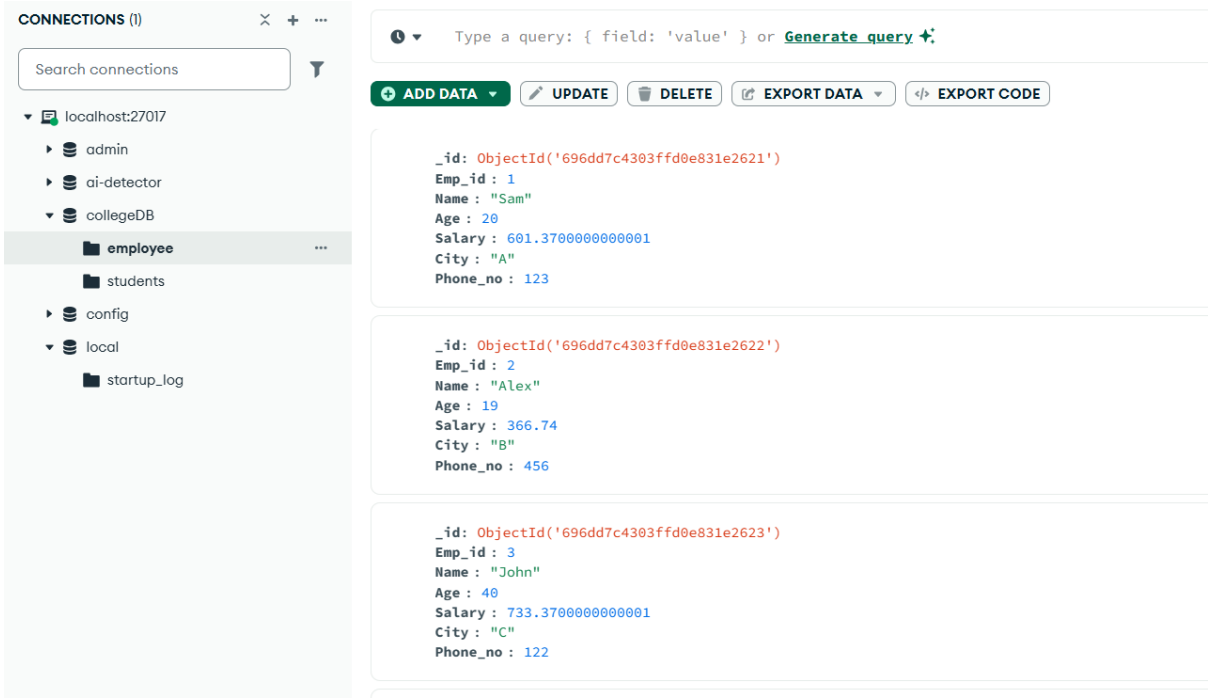
```
collegeDB> db.employee.find({ Age: 20 })
[
  {
    _id: ObjectId('696dd7c4303ffd0e831e2621'),
    Emp_id: 1,
    Name: 'Sam',
    Age: 20,
    Salary: 601.3700000000001,
    City: 'A',
    Phone_no: 123
  }
]
```

Q12. Write a query to display all the documents with AGE not equals to 20.

```
collegeDB> db.employee.find({ Age: { $ne: 20 } })
[
  {
    _id: ObjectId('696dd7c4303ffd0e831e2622'),
    Emp_id: 2,
    Name: 'Alex',
    Age: 19,
    Salary: 366.74,
    City: 'B',
    Phone_no: 456
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2623'),
    Emp_id: 3,
    Name: 'John',
    Age: 40,
    Salary: 733.3700000000001,
    City: 'C',
    Phone_no: 122
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2624'),
    Emp_id: 4,
    Name: 'Bob',
    Age: 55,
    Salary: 746.24,
    City: 'D',
    Phone_no: 444
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2626'),
    Emp_id: 6,
    Name: 'Danny',
    Age: 28,
    Salary: 600.93,
    City: 'F',
    Phone_no: 892
  }
]
```

Q13. Write a query to display all the documents where AGE is greater than equals to 30.

```
collegeDB> db.employee.find({ Age: { $gte: 30 } })
[
  {
    _id: ObjectId('696dd7c4303ffd0e831e2623'),
    Emp_id: 3,
    Name: 'John',
    Age: 40,
    Salary: 733.3700000000001,
    City: 'C',
    Phone_no: 122
  },
  {
    _id: ObjectId('696dd7c4303ffd0e831e2624'),
    Emp_id: 4,
    Name: 'Bob',
    Age: 55,
    Salary: 746.24,
    City: 'D',
    Phone_no: 444
  }
]
```



CONNECTIONS (1)

Search connections

- localhost:27017
  - admin
  - ai-detector
  - collegeDB
    - employee
    - students
  - config
  - local
    - startup_log

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT DATA | EXPORT CODE

```
Phone_no : 456
```

```
_id: ObjectId('696dd7c4303ffd0e831e2623')
Emp_id : 3
Name : "John"
Age : 40
Salary : 733.3700000000001
City : "C"
Phone_no : 122
```

```
_id: ObjectId('696dd7c4303ffd0e831e2624')
Emp_id : 4
Name : "Bob"
Age : 55
Salary : 746.24
City : "D"
Phone_no : 444
```

```
_id: ObjectId('696dd7c4303ffd0e831e2626')
Emp_id : 6
Name : "Danny"
Age : 28
Salary : 600.93
City : "F"
Phone_no : 892
```